

Lab-2 - Introduction to Packet Tracer, Building a single-segment Network, Protocol Analysis using OSI and TCP/IP Model

Objectives

- To introduce **Packet Tracer (PT)** and become familiar with its operations.
- Create a simple network using **Hub, Switch**.
- Learn about basic components of a router.
- Configure router interfaces and learn other basic router configuration settings.
- To connect hosts in different networks using a Router in Packet Tracer.

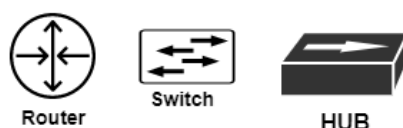
Part-A: Introduction to Packet Tracer

Part-1: Network Devices Symbol

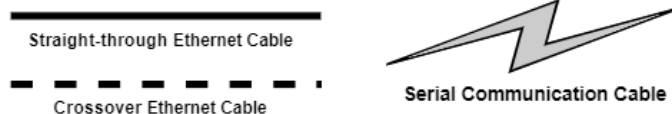
End Devices:



Intermediate Devices:



Media:

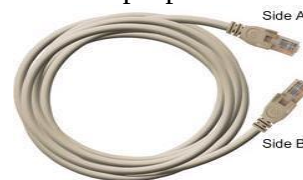


Part-2: Crossover and Straight-through Cables

Common Ethernet network cables are straight and crossover cable. This Ethernet network cable is made of 4 pair high performance cable that consists of twisted pair conductors that used for data transmission. Both end of cable is called RJ45 connector. Straight and crossover cable can be Cat3, Cat 5, Cat 5e or Cat 6 UTP cable, the only difference is each type will have different wire arrangement in the cable for serving different purposes.

Category 1 (Layer 2 or below): Hub Switch

Category 2 (Layer 3 or above): PC/Laptop/Server Router



You usually use straight cable to connect different type of devices (different category).

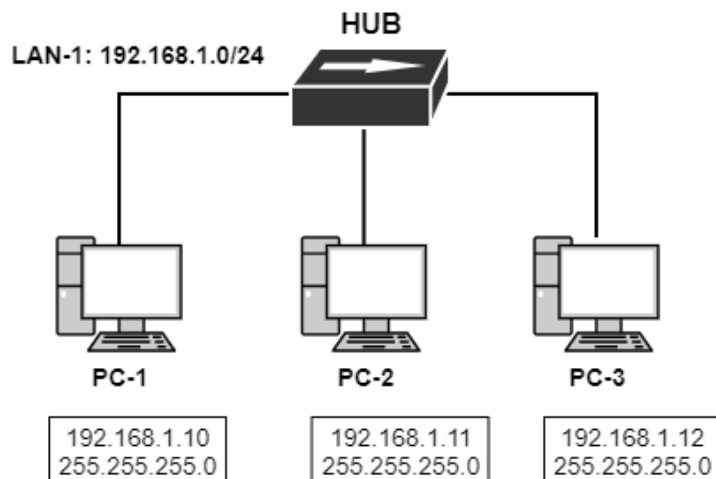
This type of cable will be used most of the time and can be used to:

- Connect a computer to a switch/hub's normal port.
- Connect a router's LAN port to a switch/hub's uplink port. (normally used for expanding network)

Sometimes, **you will use crossover cable, it's usually used to connect same type of devices or same category.** A crossover cable can be used to:

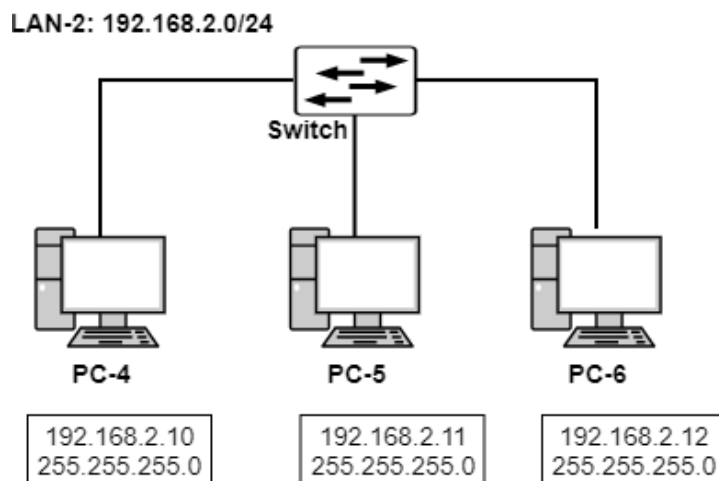
- Connect 2 computers directly.
- Connect a router's LAN port to a switch/hub's normal port. (normally used for expanding network)
- Connect 2 switches/hubs by using normal port in both switches/hubs.

Part-3: Create a single-segment network using a Hub



Test the network, e.g., ping PC2 from PC1 and ping PC2 from PC3.
Run the test in Simulation mode of Packet Tracer and Ping again

Part-4: Create a single-segment network using a Switch and Compare with the Part-3



Test the network, e.g., ping PC5 from PC4 and ping PC6 from PC5.
Run the test in Simulation mode of Packet Tracer and Ping again

Based on Part-3 and Part-4,

Which type of network is good for the enterprise solution? (Explain)

Part-B: Basic Router Configuration

Part-1: Theoretical background

Cisco Internet Operating System (IOS) Command Interface User Levels The following table contains the different IOS command modes, their roles and the shape of the command prompt that illustrates the mode. Make sure to study this table carefully as it is essential for proper working with Cisco routers and switches.

IOS command mode	Functions/Roles of the mode	Command prompt
User EXEC mode	* Limited command set, e.g., ping, telnet * Change of system parameters	Router> Router>enable Router#
Privileged EXEC mode	* Manage configuration files * Examine state of router * Access control with password	Router# Router# configure terminal Router(config)#
Global Configuration Mode	* Change system parameters	Router(config)#interface fa0/0 Router(config-if)#
Interface Mode Or, Other Configuration Mode	* Interface configuration mode: Modify configuration of a specific interface * Router configuration mode: Modify configuration of specific routing protocol	Router(config-if)# Router(config-router)#
Use TAB and “?” mark frequently for command		

Part-2: Configure a Cisco IOS Router/Switch.

PT Terminal is a simple emulation program for serial communication that can be used to connect to the console port on Cisco IOS devices. A serial interface on a computer is connected to the Cisco device via a console cable. Using PT Terminal is the most basic way to access a router for checking or changing its configuration.



Step-1: Select a PC and a router from the Network Component Box.

Step-2: Connect the console (rollover) cable to the console port on the router. Connect the other cable end to the host computer with a **RS-232 port**.

Step-3: From the Windows taskbar, start the PT Terminal program by clicking

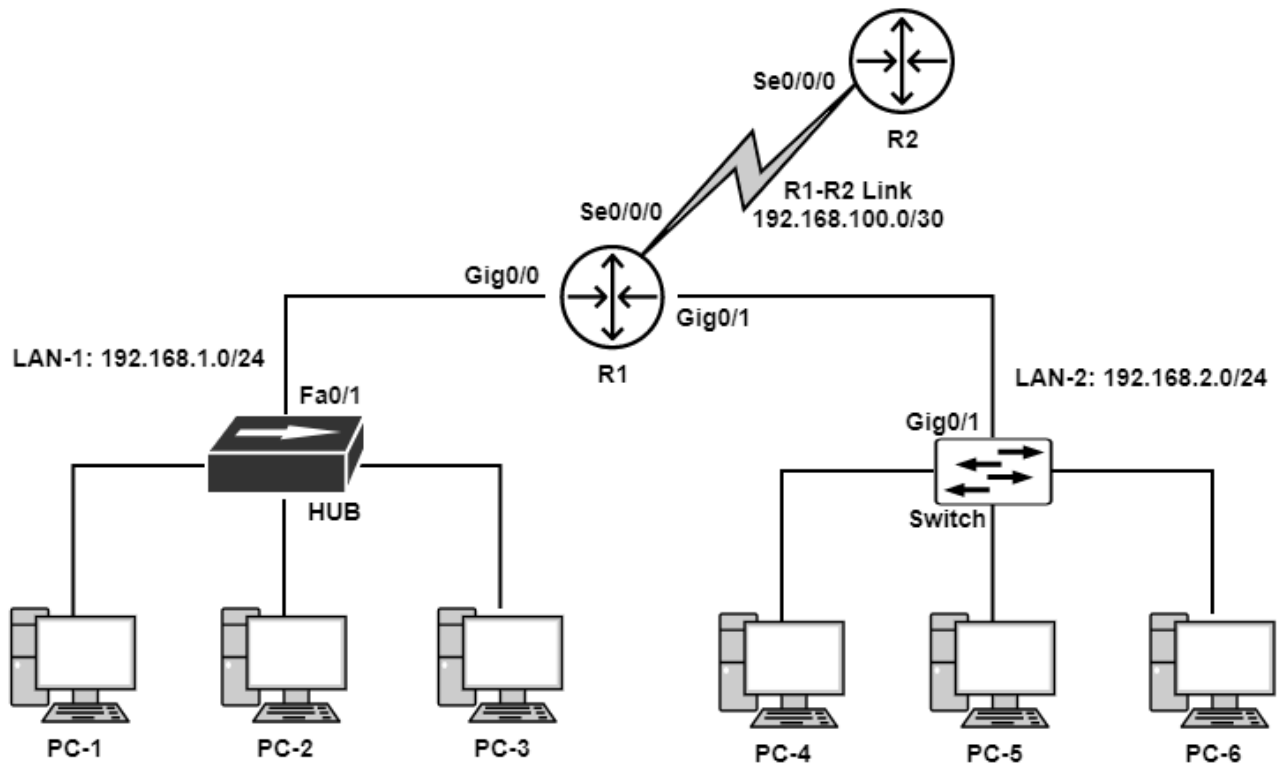
PC>Desktop Tab> Terminal

Step-4: **Click OK.**

5-You should see a response from the router on the screen (press enter several times).

Note: this process is the same for a Switch.

Part-3: Creating a Local Area Network (LAN) using Hub/Switch/Router



Part-3(A): Router console session and Global Configuration Mode

At First you need to create a console session for configure the Router. Then access via Console session and enter the specific command,

Router Console Session and User Exec mode

```
Router >
```

Router Exec mode to privileged Exec mode

```
Router >
```

```
Router >enable
```

```
Router #
```

Router privileged mode to configuration modes.

```
Router #
```

```
Router # configure terminal
```

```
Router (config) #
```

Part-3(B): Change Router Host Name as R1

```
Router (config) #
```

```
Router (config) # hostname R1
```

```
R1 (config) #
```

Part-3(C): Assign R1 Gig 0/0 IP Address and state up the interface

G0/0 ip address is assign by LAN-1 Network and G0/0 is the default gateway of the LAN-1. For design purpose, we use 1st or Last IP address as a Default Gateway.

So, we use 1st IP Address of LAN-1. IP Address: 192.168.1.1 Subnet Mask: 255.255.255.0

```
R1 (config) # interface Gigabitethernet 0/0
R1 (config-if) # ip address 192.168.1.1 255.255.255.0
R1 (config-if) # no shutdown
R1 (config-if) # exit
R1 (config) #
```

Note: Now you configure the Gig0/1 interface.

Part-3(D): Configure R1-R2 Link Network

We take 1st IP address in R1 Side,

```
R1 (config) # interface Se0/0/0
R1 (config-if) # ip address 192.168.100.1 255.255.255.252
R1 (config-if) # no shutdown
R1 (config-if) # exit
R1 (config) #
```

Note: Now you configure the R2 Serial interface.

Part-3(E): Assign End Devices IP Address with the Default Gateway.

PC-1 → Desktop>IP Configuration

IP Address: 192.168.1.10

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.1.1

Close the PC-1 window

Note: Configure the other PC's

Part-3(F): Verify network connectivity.

Use the ping command to verify network connectivity with the router. If ping replies are not successful troubleshoot the connection:

Part-3(G): Verify the running configuration and save this configuration to NVRAM

R1#show startup configuration

--- more ---

R1#show running-config

--- more ---

R1# copy running-config startup-config

Destination filename [startup-config]? <ENTER>

Building configuration...

[OK]

Part-4: Write the Answer

- What are the four primitive modes of Router IOS? What are the purposes of each mode?
- Why we use “TAB” and “?” symbol during router configuration?
- What is the basic difference between “startup configuration” file and “running configuration” file?
- What is the output of “show run” command?
- Why and in which mode “copy run start” command is used?
- Why “no shutdown” command is used?